| | | |
|---|---|---|
| GEGN580 | APPLIED REMOTE SENSING FOR GEOENGINEERING AND GEOSCIENCES | 3.0 |
| GEGN588 | ADVANCED PLANETARY GEOGRAPHIC INFORMATION SYSTEMS | 3.0 |
| GEOL557 | EARTH RESOURCE DATA SCIENCE 1: FUNDAMENTALS | 3.0 |
| GEOL558 | EARTH RESOURCE DATA SCIENCE 2: APPLICATIONS AND MACHINE-LEARNING | 3.0 |
| MNGN502 | GEOSPATIAL BIG DATA ANALYTICS | 3.0 |
| MNGN557 | MINERAL ECONOMICS AND POLICY | 2.0 |
| MNGN561 | PROJECT MANAGEMENT | 3.0 |
| MNGN562 | MINING ENVIRONMENTAL AND SOCIAL RESPONSIBILITY | 2.0 |
| DSCI503 | ADVANCED DATA SCIENCE | 3.0 |
| DSCI570 | INTRODUCTION TO MACHINE LEARNING | 3.0 |
| MATH530 | INTRODUCTION TO STATISTICAL METHODS | 3.0 |
| CEEN501 | LIFE CYCLE ASSESSMENT | 3.0 |
| Students attending on campus also may select from semester based, in person courses GEGN532, GEGN568, CEEN581. | | |

## Mines Combined Undergraduate / Graduate Program

Students enrolled in Mines' combined undergraduate/graduate program may double count up to six credits of graduate coursework to fulfill requirements of both their undergraduate and graduate degree programs. These courses must have been passed with "B-" or better, not be substitutes for required coursework, and meet all other University, Department, and Program requirements for graduate credit.

Students are advised to consult with their undergraduate and graduate advisors for appropriate courses to double count upon admission to the combined program.

**The Graduate Certificate Program in GIS & GeoInformatics** outlined below may be completed by individuals already holding an undergraduate or advanced degree or as a combined degree program by individuals already matriculated as graduate students at Colorado School of Mines. The graduate certificate is comprised of:

| | |
|---|---|
| Course Work | 12.0 |
| **Total Semester Hrs** | **12.0** |

## Graduate Certificate of GIS & GeoInformatics:

Students working towards a Graduate Certificate of GIS & GeoInformatics are required to take any four of the following courses:

| | | |
|---|---|---|
| One Required Course: | | |
| GEGN575 | APPLICATIONS OF GEOGRAPHIC INFORMATION SYSTEMS | 3.0 |
| And, any three of the following courses: | | |
| GEGN542 | ADVANCED DIGITAL TERRAIN ANALYSIS | 3.0 |
| GEGN579 | PYTHON SCRIPTING FOR GEOGRAPHIC INFORMATION SYSTEMS | 3.0 |
| GEGN580 | APPLIED REMOTE SENSING FOR GEOENGINEERING AND GEOSCIENCES | 3.0 |
| GEGN588 | ADVANCED PLANETARY GEOGRAPHIC INFORMATION SYSTEMS | 3.0 |
| GEOL557 | EARTH RESOURCE DATA SCIENCE 1: FUNDAMENTALS | 3.0 |
| GEOL558 | EARTH RESOURCE DATA SCIENCE 2: APPLICATIONS AND MACHINE-LEARNING | 3.0 |
| CEEN501 | LIFE CYCLE ASSESSMENT | 3.0 |
| DSCI503 | ADVANCED DATA SCIENCE | 3.0 |
| DSCI570 | INTRODUCTION TO MACHINE LEARNING | |
| MATH530 | INTRODUCTION TO STATISTICAL METHODS | 3.0 |
| MNGN502 | GEOSPATIAL BIG DATA ANALYTICS | 3.0 |
| MNGN557 | MINERAL ECONOMICS AND POLICY | 2.0 |
| MNGN561 | PROJECT MANAGEMENT | 3.0 |
| MNGN562 | MINING ENVIRONMENTAL AND SOCIAL RESPONSIBILITY | 2.0 |
| Students attending on campus also may select from semester based, in person courses GEGN532, GEGN568, CEEN581. | | |

### Program Director

Wendy Zhou, Professor, Geology & Geological Engineering

### Program Associate Director

Sebnem Duzgun, Professor, Fred Banfield Endowed Chair in Mining Engineering

### Professors

Sebnem Duzgun, Professor, Fred Banfield Endowed Chair in Mining Engineering

Amy Landis, Civil and Environmental Engineering, Professor, Mines Presidential Faculty Fellow for Access, Attainment & Diversity

Wendy Zhou, Professor, Geology & Geological Engineering

### Teaching Professor

Wendy Fisher, Computer Science, Teaching Professor

### Research Professor

Zane Jobe, Research Associate Professor, Chevron Center of Research Excellence Director, Geology & Geological Engineering

# Humanitarian Engineering and Science

## Program Description

The MS degrees in Humanitarian Engineering and Science (HES) are a professional MS (non-thesis) and a thesis-based MS. These degrees are targeted to recent graduates or mid-career professionals with a BS in science and engineering who are interested in careers, research opportunities, and/or acquiring skills that will help them work effectively with communities. The degrees include a core HES curriculum plus an approved track of related courses in a science or engineering discipline.

The HES graduate certificate is designed for professionals seeking to attend school part time or students who are seeking degrees in other

departments at Mines but still desire graduate training in humanitarian engineering and science. It consists of four courses.

In both the master's degrees and graduate certificates, a unique mix of social science, applied science, and engineering perspectives prepares students to apply knowledge about the earth to promote more sustainable and just uses of water, energy, and other earth resources and to understand and mitigate potential hazards.

## Primary Contact

Program Director: Richard Krahenbuhl
Program Coordinator: Rachel McDonald
humanitarian.mines.edu/mshes/

## Graduate Certificate Program Requirements

The *Humanitarian Engineering and Science (HES)* certificate is an online or residential program designed for working professionals as well as graduate students who are enrolled in other degrees at Mines but wish to gain knowledge in humanitarian engineering and science. To obtain a graduate certificate, students must complete a minimum of 9 credits of the following courses. Students may not double-count courses from their undergraduate degrees. Students who have already taken one of the classes as undergraduates must find a suitable replacement, to be approved by the HES director. Students are encouraged to take 12 credits of coursework if possible, adding an elective from the approved HES electives list below.

Required HES certificate courses (9 credits):

| | | |
|---|---|---|
| EDNS515 | INTRODUCTION TO SCIENCE AND TECHNOLOGY STUDIES | 3.0 |
| EDNS577 | ADVANCED ENGINEERING AND SUSTAINABLE COMMUNITY DEVELOPMENT | 3.0 |
| EDNS579 | COMMUNITY-BASED RESEARCH METHODS | 3.0 |
| **Total Semester Hrs** | | **9.0** |

**Master of Science (MS) Program Requirements**

The MS degrees in *Humanitarian Engineering and Science (HES)* are a professional MS (non-thesis) and a thesis-based MS. These degrees are targeted to recent graduates or midcareer professionals with a BS in science and engineering who are interested in careers, research opportunities, and/or acquiring skills that will help them work effectively with communities. The degrees include a core HES curriculum plus an approved track of related courses in a science or engineering discipline. A unique mix of social science, applied science, and engineering perspectives prepares students to apply knowledge about the earth to promote more sustainable and just uses of water, energy, and other earth resources and to understand and mitigate potential hazards.

## Master of Science (non-thesis)

To obtain the 30 credits required for the MS (non-thesis), students must satisfy the following program requirements: (1) 12 credits of required HES courses, 2) 3 credits of elective HES courses approved by Engineering, Design & Society, and 3) 15 credits of courses approved by the affiliated department (see the six tracks detailed below).

***HES MS (Non-Thesis) Core Courses (15 credits):***

| | | |
|---|---|---|
| EDNS515 | INTRODUCTION TO SCIENCE AND TECHNOLOGY STUDIES | 3.0 |
| EDNS577 | ADVANCED ENGINEERING AND SUSTAINABLE COMMUNITY DEVELOPMENT | 3.0 |
| EDNS579 | COMMUNITY-BASED RESEARCH METHODS | 3.0 |
| EDNS580 | HUMANITARIAN ENGINEERING AND SCIENCE CAPSTONE PRACTICUM | 3.0 |
| ELECTIVE | An approved HES elective from the list below | 3.0 |
| **Total Semester Hrs** | | **15.0** |

***Approved HES Electives:***

| | | |
|---|---|---|
| EDNS590 | RISKS IN HUMANITARIAN ENGINEERING AND SCIENCE | 3.0 |
| CEEN501 | LIFE CYCLE ASSESSMENT | 3.0 |
| CEEN575 | HAZARDOUS WASTE SITE REMEDIATION | 3.0 |
| CEEN593 | SUSTAINABLE ENGINEERING DESIGN | 3.0 |
| CEEN556 | MINING AND THE ENVIRONMENT | 3.0 |
| CEEN570 | WATER AND WASTEWATER TREATMENT | 3.0 |
| CEEN573 | RECLAMATION OF DISTURBED LANDS | 3.0 |
| CEEN580 | CHEMICAL FATE AND TRANSPORT IN THE ENVIRONMENT | 3.0 |
| CEEN581 | WATERSHED SYSTEMS MODELING | 3.0 |
| CEEN595 | ANALYSIS OF ENVIRONMENTAL IMPACT | 3.0 |
| EBGN553 | PROJECT MANAGEMENT | 3.0 |
| HASS525 | ENVIRONMENTAL COMMUNICATION | 3.0 |
| HASS526 | INTERCULTURAL COMMUNICATION | 3.0 |
| HASS527 | RISK COMMUNICATION | 3.0 |
| HASS565 | SCIENCE, TECHNOLOGY, AND SOCIETY | 3.0 |
| HASS568 | ENVIRONMENTAL JUSTICE | 3.0 |
| HASS590 | ENERGY AND SOCIETY | 3.0 |
| MNGN503 | MINING TECHNOLOGY FOR SUSTAINABLE DEVELOPMENT | 3.0 |
| MNGN510 | FUNDAMENTALS OF MINING AND MINERAL RESOURCE DEVELOPMENT | 3.0 |
| MNGN565 | MINE RISK MANAGEMENT | 3.0 |
| MNGN567 | SUSTAINABLE DEVELOPMENT AND EARTH RESOURCES | 3.0 |
| MNGN571 | ENERGY, NATURAL RESOURCES, AND SOCIETY | 3.0 |
| PEGN530 | ENVIRONMENTAL LAW AND SUSTAINABILITY | 3.0 |

## Disciplinary Tracks

***Track 1: Geophysics (GPGN) (15 credits):***

Degree candidates should have an undergraduate degree in geophysics, physics, quantitative earth sciences and engineering, or equivalent coursework. In addition, candidates will need to complete necessary prerequisite courses for the graduate courses.

In addition to the core HES MS (non-thesis) curriculum (15 credits) detailed above, MS (non-thesis) students following the Geophysics track must take one required course (3 credits) and at least 12 credits of approved elective courses, as shown below. Courses not listed below that align with the student's practicum can be substituted in consultation with the degree advisor.

| **Required Course** | | |
|---|---|---|
| GPGN577 | HUMANITARIAN GEOSCIENCE | 3.0 |

| | | |
|---|---|---|
| **At least four courses of the following:** | | |
| GPGN520 | ELECTRICAL AND ELECTROMAGNETIC EXPLORATION | 3.0 |
| GPGN570 | APPLICATIONS OF SATELLITE REMOTE SENSING | 3.0 |
| GPGN574 | ADVANCED HYDROGEOPHYSICS | 3.0 |
| GPGN590 | INSTRUMENTAL DESIGN IN APPLIED GEOSCIENCES | 3.0 |
| GEGN532 | GEOLOGICAL DATA ANALYSIS | 3.0 |

### *Track 2: Environmental Engineering (CEEN) (15 credits):*

A BS degree in a science or engineering discipline is required. Pre-requisites include two semesters of college calculus, one semester of college physics, two semesters of college chemistry, and one semester of college statistics.

In addition to the core HES MS (non-thesis) curriculum (15 credits) detailed above, MS (non-thesis) students following the Environmental Engineering track must take three required courses (9 credits) and at least two courses (6 credits) of approved elective courses, as shown below. Courses not listed below that align with the student's practicum can be substituted in consultation with the degree advisor.

| | | |
|---|---|---|
| **Required Courses:** | | |
| GPGN577 | HUMANITARIAN GEOSCIENCE | 3.0 |
| CEEN550 | PRINCIPLES OF ENVIRONMENTAL CHEMISTRY | 3.0 |
| CEEN580 | CHEMICAL FATE AND TRANSPORT IN THE ENVIRONMENT | 3.0 |
| **At least two courses of the following:** | | |
| **Environmental Microbiology** | | |
| CEEN560 | MOLECULAR MICROBIAL ECOLOGY AND THE ENVIRONMENT | 3.0 |
| CEEN562 | ENVIRONMENTAL GEOMICROBIOLOGY | 3.0 |
| CEEN566 | MICROBIAL PROCESSES, ANALYSIS AND MODELING | 3.0 |
| **Treatment** | | |
| CEEN570 | WATER AND WASTEWATER TREATMENT | 3.0 |
| CEEN575 | HAZARDOUS WASTE SITE REMEDIATION | 3.0 |
| MNGN556 | MINE WATER AND ENVIRONMENT | 3.0 |
| **Hydrology** | | |
| CEEN555 | LIMNOLOGY | 3.0 |
| CEEN581 | WATERSHED SYSTEMS MODELING | 3.0 |
| GEGN582 | INTEGRATED SURFACE WATER HYDROLOGY | 3.0 |
| GEGN584 | FIELD METHODS IN HYDROLOGY | 3.0 |

### *Track 3: Geological Engineering (GEGN) (15 credits):*

Degree candidates should have an undergraduate degree in engineering or the equivalent coursework. In addition, candidates will need to complete necessary prerequisite courses for the graduate courses, including engineering geology, groundwater engineering, soil mechanics, and rock mechanics.

In addition to the core HES MS (non-thesis) curriculum (15 credits) detailed above, MS (non-thesis) students following the Geological Engineering track must take two required courses (6 credits) and at least three courses (9 credits) of approved elective courses, as shown below.

| | | |
|---|---|---|
| **Required Courses:** | | |
| GEGN532 | GEOLOGICAL DATA ANALYSIS | 3.0 |
| GPGN577 | HUMANITARIAN GEOSCIENCE | 3.0 |
| **Candidates must also take at least three of the following courses. The student and the instructor will work together to develop humanitarian themes in the project assignments within each course.** | | |
| GEGN563 | APPLIED NUMERICAL MODELLING FOR GEOMECHANICS | 3.0 |
| GEGN570 | CASE HISTORIES IN GEOLOGICAL ENGINEERING AND HYDROGEOLOGY | 3.0 |
| GEGN573 | GEOLOGICAL ENGINEERING SITE INVESTIGATION | 3.0 |
| GEGN575 | APPLICATIONS OF GEOGRAPHIC INFORMATION SYSTEMS | 3.0 |
| GEGN580 | APPLIED REMOTE SENSING FOR GEOENGINEERING AND GEOSCIENCES | 3.0 |
| GEGN671 | LANDSLIDES: INVESTIGATION, ANALYSIS & MITIGATION | 3.0 |
| GEGN673 | ADVANCED GEOLOGICAL ENGINEERING DESIGN | 3.0 |

### *TRACK 4: Humanitarian Robotics (15 CREDITS):*

Degree candidates should have an undergraduate degree in computer science, mechanical or electrical engineering, or robotics, or equivalent coursework. In addition, candidates will need to complete necessary prerequisite courses for the graduate courses.

In addition to the Core HES MS (non-thesis) curriculum (15 credits) detailed above, MS (non-thesis) students following the Humanitarian Robotics track must take three required course (9 credits) and at least 6 credits of approved elective courses, as shown below. Courses not listed below that align with the student's practicum can be substituted in consultation with the degree advisor.

| | | |
|---|---|---|
| **Required Courses:** | | |
| CSCI532 | ROBOT ETHICS | 3.0 |
| CSCI536 | HUMAN-ROBOT INTERACTION | 3.0 |
| CSCI573 | ROBOT PROGRAMMING AND PERCEPTION | 3.0 |
| **At least two courses from the following:** | | |
| CSCI507 | INTRODUCTION TO COMPUTER VISION | 3.0 |
| CSCI534 | ROBOT PLANNING AND MANIPULATION | 3.0 |
| CSCI575 | ADVANCED MACHINE LEARNING | 3.0 |
| EENG517 | THEORY AND DESIGN OF ADVANCED CONTROL SYSTEMS | 3.0 |
| EENG519 | ESTIMATION THEORY AND KALMAN FILTERING | 3.0 |
| MEGN540 | MECHATRONICS | 3.0 |
| MEGN544 | ROBOT MECHANICS: KINEMATICS, DYNAMICS, AND CONTROL | 3.0 |
| MEGN545 | ADVANCED ROBOT CONTROL | 3.0 |

### *TRACK 5: Data Science (DSCI) (15 CREDITS):*

Degree candidates should have an undergraduate degree in computer science, mathematics, or data science or equivalent coursework. In addition, candidates will need to complete necessary prerequisite courses for the graduate courses.

In addition to the core HES MS (non-thesis) curriculum (15 credits) detailed above, MS (non-thesis) students following the Data Science track must take four required courses (12 credits) and at least 3 credits of approved elective courses, as shown below. In addition to earning the HES MS (non-thesis) degree, they will also earn the Data Science Statistical Learning Graduate Certificate.

| **Required Courses** | | |
|---|---|---|
| DSCI503 | ADVANCED DATA SCIENCE | 3.0 |
| MATH530 | INTRODUCTION TO STATISTICAL METHODS | 3.0 |
| MATH560 | INTRODUCTION TO KEY STATISTICAL LEARNING METHODS I | 3.0 |
| MATH561 | INTRODUCTION TO KEY STATISTICAL LEARNING METHODS II | 3.0 |
| **At least one course of the following:** | | |
| MATH532 | SPATIAL STATISTICS | 3.0 |
| MATH533 | TIME SERIES ANALYSIS AND ITS APPLICATIONS | |
| MATH536 | ADVANCED STATISTICAL MODELING | 3.0 |
| MATH537 | MULTIVARIATE ANALYSIS | 3.0 |
| MATH582 | STATISTICS PRACTICUM | 3.0 |

### *TRACK 6: Interdisciplinary (15 CREDITS):*

In addition to the core HES MS (non-thesis) curriculum (15 credits) detailed above, MS (non-thesis) students following the Interdisciplinary track will work with their advisor to choose an additional 15 credits that best match their intellectual interests. As with our other tracks, at least 12 of these credits need to be engineering or applied science courses.

Students seeking this track are required to identify their desired focus area when applying and identify possible courses upon matriculation. They will then work with their advisor to ensure that the student meets the course prerequisites and that the courses are offered on an appropriate timetable according to their anticipated graduation date.

## Master of Science (thesis)

To obtain the 30 credits required for the MS (thesis), students must satisfy the following program requirements: 1) 9 credits of required HES core courses, 2) 3 credits of elective HES classes approved by Engineering, Design and Society, 3) 12 credits of approved Disciplinary Track classes, and 4) 6 credits of MS thesis research on a thesis topic approved by HES faculty in the Engineering, Design, and Society Division and the affiliated disciplinary track.

***HES MS (thesis) Core Courses (12 credits):***

| | | |
|---|---|---|
| EDNS515 | INTRODUCTION TO SCIENCE AND TECHNOLOGY STUDIES | 3.0 |
| EDNS577 | ADVANCED ENGINEERING AND SUSTAINABLE COMMUNITY DEVELOPMENT | 3.0 |
| EDNS579 | COMMUNITY-BASED RESEARCH METHODS | 3.0 |
| ELECTIVE | 3 credits of approved HES electives from list below | 3.0 |

***Approved HES Electives:***

| | | |
|---|---|---|
| EDNS590 | RISKS IN HUMANITARIAN ENGINEERING AND SCIENCE | 3.0 |
| CEEN501 | LIFE CYCLE ASSESSMENT | 3.0 |
| CEEN575 | HAZARDOUS WASTE SITE REMEDIATION | 3.0 |
| CEEN593 | SUSTAINABLE ENGINEERING DESIGN | 3.0 |
| CEEN556 | MINING AND THE ENVIRONMENT | 3.0 |
| CEEN570 | WATER AND WASTEWATER TREATMENT | 3.0 |
| CEEN573 | RECLAMATION OF DISTURBED LANDS | 3.0 |
| CEEN580 | CHEMICAL FATE AND TRANSPORT IN THE ENVIRONMENT | 3.0 |
| CEEN581 | WATERSHED SYSTEMS MODELING | 3.0 |
| CEEN595 | ANALYSIS OF ENVIRONMENTAL IMPACT | 3.0 |
| EBGN553 | PROJECT MANAGEMENT | 3.0 |
| HASS525 | ENVIRONMENTAL COMMUNICATION | 3.0 |
| HASS526 | INTERCULTURAL COMMUNICATION | 3.0 |
| HASS527 | RISK COMMUNICATION | 3.0 |
| HASS565 | SCIENCE, TECHNOLOGY, AND SOCIETY | 3.0 |
| HASS568 | ENVIRONMENTAL JUSTICE | 3.0 |
| HASS590 | ENERGY AND SOCIETY | 3.0 |
| MNGN503 | MINING TECHNOLOGY FOR SUSTAINABLE DEVELOPMENT | 3.0 |
| MNGN510 | FUNDAMENTALS OF MINING AND MINERAL RESOURCE DEVELOPMENT | 3.0 |
| MNGN565 | MINE RISK MANAGEMENT | 3.0 |
| MNGN567 | SUSTAINABLE DEVELOPMENT AND EARTH RESOURCES | 3.0 |
| MNGN571 | ENERGY, NATURAL RESOURCES, AND SOCIETY | 3.0 |
| PEGN530 | ENVIRONMENTAL LAW AND SUSTAINABILITY | 3.0 |

## Disciplinary Tracks

### *Track 1: Geophysics (GPGN) Courses and Thesis (18 credits):*

Degree candidates should have an undergraduate degree in geophysics, physics, quantitative earth sciences, or engineering or equivalent coursework. In addition, candidates will need to complete necessary prerequisite courses for the graduate courses.

In addition to the core HES MS (thesis) curriculum (12 credits) detailed above, MS (thesis) students following the Geophysics track must take one required course (3 credits), at least 9 credits of approved elective courses, and 6 credits of independent thesis research, as shown below.

Courses not listed below that align with the student's thesis can be substituted in consultation with the degree advisor.

| **Required Course** | | |
|---|---|---|
| GPGN577 | HUMANITARIAN GEOSCIENCE | 3.0 |
| **At least three courses of the following:** | | |

| | | |
|---|---|---|
| GPGN520 | ELECTRICAL AND ELECTROMAGNETIC EXPLORATION | 3.0 |
| GPGN570 | APPLICATIONS OF SATELLITE REMOTE SENSING | 3.0 |
| GPGN574 | ADVANCED HYDROGEOPHYSICS | 3.0 |
| GPGN590 | INSTRUMENTAL DESIGN IN APPLIED GEOSCIENCES | 3.0 |
| GEGN532 | GEOLOGICAL DATA ANALYSIS | 3.0 |
| **And:** | | |
| GPGN707 | GRADUATE THESIS / DISSERTATION RESEARCH CREDIT | 6.0 |

### *Track 2: Environmental Engineering (CEEN) (18 credits):*

A BS degree in a science or engineering discipline is required. Pre-requisites include two semesters of college calculus, one semester of college physics, two semesters of college chemistry, and one semester of college statistics.

In addition to the Core HES MS (thesis) curriculum (12 credits) detailed above, MS (thesis) students following the Environmental Engineering track must take one required course (3 credits), at least two courses (6 credits) of approved elective courses, and 6 credits of independent thesis research, as shown below. Courses not listed below that align with the student's thesis can be substituted in consultation with the degree advisor.

| | | |
|---|---|---|
| **Required Course:** | | |
| GPGN577 | HUMANITARIAN GEOSCIENCE | 3.0 |
| **At least three courses of the following:** | | |
| CEEN550 | PRINCIPLES OF ENVIRONMENTAL CHEMISTRY | 3.0 |
| CEEN580 | CHEMICAL FATE AND TRANSPORT IN THE ENVIRONMENT | 3.0 |
| **Environmental Microbiology** | | |
| CEEN560 | MOLECULAR MICROBIAL ECOLOGY AND THE ENVIRONMENT | 3.0 |
| CEEN562 | ENVIRONMENTAL GEOMICROBIOLOGY | 3.0 |
| CEEN566 | MICROBIAL PROCESSES, ANALYSIS AND MODELING | 3.0 |
| **Treatment** | | |
| CEEN570 | WATER AND WASTEWATER TREATMENT | 3.0 |
| CEEN575 | HAZARDOUS WASTE SITE REMEDIATION | 3.0 |
| MNGN556 | MINE WATER AND ENVIRONMENT | 3.0 |
| **Hydrology** | | |
| CEEN555 | LIMNOLOGY | 3.0 |
| CEEN581 | WATERSHED SYSTEMS MODELING | 3.0 |
| GEGN582 | INTEGRATED SURFACE WATER HYDROLOGY | 3.0 |
| GEGN584 | FIELD METHODS IN HYDROLOGY | 3.0 |
| **And** | | |
| CEEN707 | GRADUATE THESIS / DISSERTATION RESEARCH CREDIT | 6.0 |

### *Track 3: Geological Engineering (GEGN) (18 credits):*

Degree candidates should have an undergraduate degree in engineering or the equivalent coursework. In addition, candidates will need to complete necessary prerequisite courses for the graduate courses, including engineering geology, groundwater engineering, soil mechanics, and rock mechanics.

In addition to the Core HES MS (thesis) curriculum (12 credits) detailed above, MS (thesis) students following the Geological Engineering track must take two required courses (6 credits), at least two courses (6 credits) of approved elective courses, and 6 credits of independent thesis research, as shown below.

| | | |
|---|---|---|
| **Required Course:** | | |
| GEGN532 | GEOLOGICAL DATA ANALYSIS | 3.0 |
| GPGN577 | HUMANITARIAN GEOSCIENCE | 3.0 |
| **At least two of the following courses:** | | |
| GEGN563 | APPLIED NUMERICAL MODELLING FOR GEOMECHANICS | 3.0 |
| GEGN570 | CASE HISTORIES IN GEOLOGICAL ENGINEERING AND HYDROGEOLOGY | 3.0 |
| GEGN573 | GEOLOGICAL ENGINEERING SITE INVESTIGATION | 3.0 |
| GEGN575 | APPLICATIONS OF GEOGRAPHIC INFORMATION SYSTEMS | 3.0 |
| GEGN580 | APPLIED REMOTE SENSING FOR GEOENGINEERING AND GEOSCIENCES | 3.0 |
| GEGN671 | LANDSLIDES: INVESTIGATION, ANALYSIS & MITIGATION | 3.0 |
| GEGN673 | ADVANCED GEOLOGICAL ENGINEERING DESIGN | 3.0 |
| **And:** | | |
| GEGN707 | GRADUATE THESIS / DISSERTATION RESEARCH CREDIT | 6.0 |

### *Track 4: Humanitarian Robotics (18 CREDITS):*

Degree candidates should have an undergraduate degree in computer science, mechanical or electrical engineering, or robotics or equivalent coursework. In addition, candidates will need to complete necessary prerequisite courses for the graduate courses.

In addition to the core HES MS (thesis) curriculum (12 credits) detailed above, MS (thesis) students following the Humanitarian Robotics track must take three required course (9 credits), at least 3 credits of approved elective courses, and 6 credits of independent thesis research, as shown below. Courses not listed below that align with the student's thesis can be substituted in consultation with the degree advisor.

| | | |
|---|---|---|
| **Required Courses:** | | |
| CSCI532 | ROBOT ETHICS | 3.0 |
| CSCI536 | HUMAN-ROBOT INTERACTION | 3.0 |
| CSCI573 | ROBOT PROGRAMMING AND PERCEPTION | 3.0 |
| **At least one course from the following:** | | |
| CSCI507 | INTRODUCTION TO COMPUTER VISION | 3.0 |
| CSCI534 | ROBOT PLANNING AND MANIPULATION | 3.0 |
| CSCI575 | ADVANCED MACHINE LEARNING | 3.0 |
| EENG517 | THEORY AND DESIGN OF ADVANCED CONTROL SYSTEMS | 3.0 |
| EENG519 | ESTIMATION THEORY AND KALMAN FILTERING | 3.0 |

| | | |
|---|---|---|
| MEGN540 | MECHATRONICS | 3.0 |
| MEGN544 | ROBOT MECHANICS: KINEMATICS, DYNAMICS, AND CONTROL | 3.0 |
| MEGN545 | ADVANCED ROBOT CONTROL | 3.0 |
| **And:** | | |
| CSCI707 | GRADUATE THESIS / DISSERTATION RESEARCH CREDIT | 6.0 |

### *Track 5: Data Science (DSCI) (18 CREDITS):*

Degree candidates should have an undergraduate degree in computer science, mathematics, or data science or equivalent coursework. In addition, candidates will need to complete necessary prerequisite courses for the graduate courses.

In addition to the core HES MS (thesis) curriculum (12 credits) detailed above, MS (thesis) students following the Data Science track must take four required courses (12 credits) and 6 credits of independent thesis research, as shown below. In addition to earning the HES MS (thesis) degree, they will also earn the Data Science Statistical Learning Graduate Certificate.

| **Required Courses** | | |
|---|---|---|
| DSCI503 | ADVANCED DATA SCIENCE | 3.0 |
| DSCI530 | STATISTICAL METHODS I | 3.0 |
| DSCI560 | INTRODUCTION TO KEY STATISTICAL LEARNING METHODS I | 3.0 |
| DSCI561 | INTRODUCTION TO KEY STATISTICAL LEARNING METHODS II | 3.0 |
| **And:** | | |
| MATH707 | GRADUATE THESIS / DISSERTATION RESEARCH CREDIT | 6.0 |

### *Track 6: Interdisciplinary (18 CREDITS):*

In addition to the core HES MS (thesis) curriculum (12 credits) detailed above, MS (thesis) students following the Interdisciplinary track will work with their advisor to choose an additional 12 elective credits that best match their intellectual interests and take 6 credits of independent thesis research. The 12 elective credits need to be engineering or applied science courses. Students seeking this track are required to identify their desired focus area when applying and identify possible courses upon matriculation. They will then work with their advisor to ensure that the student meets the course prerequisites and that the courses are offered on an appropriate timetable according to their anticipated graduation date.

## Mines' Combined Undergraduate/Graduate Degree Program

Students enrolled in Mines' combined undergraduate/graduate program may double count up to 6 credits of graduate coursework to fulfill requirements of both their undergraduate and graduate degree programs. These courses must have been passed with B- or better, not be substitutes for required coursework, and meet all other university, department, and program requirements for graduate credit.

Students are advised to consult with their undergraduate and graduate advisors for appropriate courses to double count upon admission to the combined program.

**EDNS479. COMMUNITY-BASED RESEARCH. 3.0 Semester Hrs.**
Engineers and applied scientists face challenges that are profoundly socio-technical in nature, and communities are increasingly calling for greater participation in the decisions that affect them. Understanding the diverse perspectives of communities and being able to establish positive working relationships with their members is therefore crucial to the socially responsible practice of engineering and applied science. This course provides students with the conceptual and methodological tools to conduct community-based research. Students will learn ethnographic field methods and participatory research strategies, and critically assess the strengths and limitations of these through a final original research project. Prerequisite: HNRS105, HNRS115 or HASS100 or graduate student standing. Co-requisite: HASS215 or graduate student standing.

**EDNS501. COMMUNITY-CENTERED APPROACHES TO RESILIENCE. 3.0 Semester Hrs.**
Community resilience to the inevitable effects of a changing climate is an emmerging topic in local and global community development spheres. This course will provide learners with a survey of how institutions from governments, donors, to public and private organizations are quickly pivoting from a world of 'sustainable development goals' to one of resilience, remediation, and adaptation at national, subnational, and local scales, and how sociotechnical perspectives can support their participation in this new world. At the end of this eight-week course, learners will be able to assess and answer the following questions: [1] what is 'community resilience' and how do different communities, stakeholders, and sectors define it, [2] how does new terminology around 'resilience' relate to previous humanitarian efforts aimed at sustainable community development, [3] what are significant sociotechnical pillars of resilience, including built infrastructure and the environment, social infrastructure and governance, [4] how can mainstream efforts aimed at resilience lead to inequality and perpetuate systemic power imbalance. Experiential learning through guest speakers and case studies will illuminate the varied and multidimensional and intersectional aspects of resilience.
**Course Learning Outcomes**

**EDNS502. MANAGING AND LEADING IN MULTISTAKEHOLDER ENVIRONMENTS. 3.0 Semester Hrs.**
Managing, leading, and dare we say 'solving' wicked public interest issues requires multistakeholder engagement across sectors, agencies, and public interest domains. This course is designed to provide learners with tools to manage and lead across these complex multistakeholder environments, and to navigate wicked problems, like climate change and community resilience. This course is designed for those who are or aspire to be public-facing managers and leaders—including those working for state and federal agencies, nonprofits, non-governmental organizations (NGOs), and quasi-governmental organizations. Key learning objectives include [1] defining the public interest, [2] assessing stakeholder environments, [3] exploring ethical decision-making tools and considering competing stakeholder interests, [4] analyzing the functions and roles of the 'responsible' public manager, [5] synthesizing modern community-engaged leadership styles and applying them in practice, [6] exploring current challenges and perspectives from leaders in relevant fields and organizations, and [6] learning and applying advanced communication skills.
**Course Learning Outcomes**

**EDNS515. INTRODUCTION TO SCIENCE AND TECHNOLOGY STUDIES. 3.0 Semester Hrs.**
This course engages scholarship on the inextricable link between science, engineering and the various social contexts within which scientists and engineers work. We begin by critically reflecting on the

question, What are science and engineering for? We then explore key conceptual domains in the social scientific study of science and engineering, including knowledge, agency, and expertise. We will learn from a diverse set of social scientific experts who study and collaborate with scientists and engineers. Students will leave the course with a better understanding of how social scientific inquiry can aid in understanding, and practicing, science and engineering. They will also have a clearer articulation of their individual professional commitments and how those fit with more traditional understandings of science and engineering.

**Course Learning Outcomes**

- By the end of this course, students will have demonstrated the ability to:
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**EDNS540. PROTOTYPING FOR CONCEPT VALIDATION. 3.0 Semester Hrs.**

This course develops students' ability to use prototypes as strategic tools for concept exploration, decision-making, and validation throughout engineering design processes. Rather than treating prototypes as final deliverables, students will conceive, model, and fabricate a series of purpose-driven prototypes to test requirements, functionality, user experience, system integration, and design for manufacturability. Through iterative design cycles, students will explore the trade-offs between materials, fidelity, and media to collect targeted data, solicit stakeholder feedback, assess market fit, and evaluate associations of comparative fabrication and assembly methods' effects on cost. Emphasis is placed on selecting appropriate prototype forms at various milestones of engineering problem solving to effectively validate hypotheses and support data-driven design decisions.

**Course Learning Outcomes**

- Conceive and deploy prototypes to verify or validate specific component or system requirements, design specifications, functions, expected user experiences, effective system integrations, and other granular aspects of a larger solution concept.
- Utilize prototypes to solicit feedback from potential customers or other stakeholders to confirm market fit and desirability.
- Explore the trade-offs between levels of fidelity and various media as they relate to data collection and concept validation.
- Evaluate fabrication methods to assess prototyping costs in relation to manufacturability and projected manufacturing costs.
- Incorporate rapid iteration cycles with minimal viable products (MVPs) to test hypotheses with real-world feedback.
- Ability to critically analyze and apply scholarship to design decision making

**EDNS544. INNOV8X. 3.0 Semester Hrs.**

Innov8x introduces concepts and tools to accelerate the design, validation and adoption of innovations in support of creative problem solving. Using an entrepreneurial mindset, we learn how to identify and frame problems that beneficiaries and stakeholders face. We attempt to design and test practical solutions to those problems in collaboration with those who experience the problems. We apply beneficiary discovery, pretotyping, business model design (social, economic and environmental), constrained creativity, efficient experimentation, and rapid iteration. While resolving challenges involves technical solutions, an important aspect of this course is directly engaging beneficiaries and stakeholders in social contexts to develop solutions with strong impact potential. Innov8x is grounded in collaborative creativity theory at the intersection of organizational behavior (social psychology), design principles, entrepreneurship and innovation management.

**Course Learning Outcomes**

- Frame and translate complex ambiguous problems into actionable opportunities for innovation
- Conduct effective, objective and ongoing beneficiary discovery in efficient ways
- Combine tools and methods to quickly test assumptions and secure beneficiary acceptance
- Develop creative approaches to navigate real and perceived constraints
- Leverage mentor and stakeholder support through credible communication based on research • Launch innovative solutions with the advocacy of beneficiaries and stakeholders
- Create value by solving complex problems that straddle technical and social domains
- Launch innovative solutions with the advocacy of beneficiaries and stakeholders

**EDNS545. PRODUCT REDESIGN. 3.0 Semester Hrs.**

Product redesign reimagines existing products, focusing specifically on a systems approach to human-centered design and the crafting of design solutions tailored to meet the needs of their users. Students will progress through an iterative design process, engaging in the analysis of and thoughtful reflection on design opportunities, ensuring enhanced products align with the needs of a specific user group. Emphasizing collaborative learning, students will work in teams, adopting a multi-disciplinary approach to creative problem-solving and design. Multiple prototyping cycles will guide students as they make data-driven design decisions, culminating in the development and communication of a final redesigned product.

**Course Learning Outcomes**

- Analyze the needs of a specific group of users in a given context and develop a problem definition that responds to those needs with a clear, concise set of engineering design criteria.
- Create a product design and development plan with a defined timeline that results in an advanced design artifact.
- Propose distinct solution concepts and utilize user feedback, engineering analysis, experimentation, and proven industry practices to make data-driven design decisions.
- Build, test, and analyze solution concepts through a series of design cycles to iterate and refine the advanced artifact.
- Participate equitably on a team with distributed roles and responsibilities, while monitoring individual effectiveness in contributing to the team's overall progress.
- Ability to critically analyze and apply scholarship to design decision making

**EDNS577. ADVANCED ENGINEERING AND SUSTAINABLE COMMUNITY DEVELOPMENT. 3.0 Semester Hrs.**

Analyzes the relationship between engineering and sustainable community development (SCD) from historical, political, ethical, cultural,

and practical perspectives. Students will study and analyze different dimensions of sustainability, development, and "helping", and the role that engineering might play in each. Will include critical explorations of strengths and limitations of dominant methods in engineering problem solving, design and research for working in SCD. Through case-studies, students will analyze and evaluate projects in SCD and develop criteria for their evaluation. 3 hours lecture and discussion; 3 semester hours.
**Course Learning Outcomes**

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**EDNS578. ENGINEERING AND SOCIAL JUSTICE. 3.0 Semester Hrs.**
Equivalent with LAIS578,
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**EDNS579. COMMUNITY-BASED RESEARCH METHODS. 3.0 Semester Hrs.**
Engineers and applied scientists face challenges that are profoundly sociotechnical in nature, and communities are increasingly calling for greater participation in the decisions that affect them. Understanding the diverse perspectives of communities and being able to establish positive working relationships with their members is therefore crucial to the socially responsible practice of engineering and applied science. This course provides graduate students with the conceptual and methodological tools to conduct community-based research. Graduate students will learn ethnographic field methods and participatory research strategies, and critically assess the strengths and limitations of these through a final original research project related to their ongoing independent research or practicums.
**Course Learning Outcomes**

- During this course students will learn to:

**EDNS580. HUMANITARIAN ENGINEERING AND SCIENCE CAPSTONE PRACTICUM. 3.0 Semester Hrs.**
(I, II, S) This course allows students to practice the concepts, theories and methods learned in HES courses with the goal of making relevant their academic training to real world problems. This practicum can be achieved through a number of possibilities approved by HES director, including supervision and/or shadowing in HES-related activities, engaging in a social enterprise where they do problem definition, impact gap analysis and layout a business canvas, and designing and carrying out a project or fieldwork of their own, etc. Prerequisite: EDNS570, EDNS479. 3 hours research; 3 semester hours.
**Course Learning Outcomes**

- • Identify successful practices for humanitarian projects in real settings (ABET a,h,j) • Determine different ways in which previous humanitarian projects could have been improved to yield more successful technical and social results (ABET a,b,h,j) • Determine effective engineering methods for different humanitarian applications (ABET b,c,h,j) • Work in teams to design, execute and evaluate a project with stakeholders (ABET a,b,c,d,e,j,k) • Gain experience in engaging and communicating with community members and stakeholders (ABET c,d,f,h,i,j,k) • Develop stronger professional communication skills through written assignments, group projects, discussions, presentations, and community engagement (ABET g,f,h,i,j,k)

**EDNS590. RISKS IN HUMANITARIAN ENGINEERING AND SCIENCE. 3.0 Semester Hrs.**
(I) This course provides students with opportunities to consider the risks related to humanitarian projects—or any projects that effect and involve people. These risks might include things that different scientific and engineering disciplines typically consider, as well as those that may be pertinent to project stakeholder perspectives. Guided by social scientific insights related to risk, students in this class will gain new tools for defining problems in ways that are relevant and appropriate for multiple contexts. Students will read, discuss, and analyze material together and to undertake independent research to deepen their understandings of chosen topics. 3 semester hours.
**Course Learning Outcomes**

- • Analyze humanitarian science and engineering projects using established evaluation criteria (ABET a,h,j) • Identify the most successful practices for humanitarian science and engineering (ABET a,h,j) • Determine different ways in which previous engineering or scientific projects could have been improved to yield more successful technical and social results (ABET a,b,h,j) • Gain conceptual tools for and experience in engaging and communicating with community members and stakeholders (ABET c,d,f,h,i,j,k) • Develop stronger professional communication skills through written assignments, group projects, discussions, presentations, and community engagement (ABET g,f,h,i,j,k)

**EDNS597. SUMMER PROGRAMS. 0-6 Semester Hr.**

**EDNS598. SPECIAL TOPICS IN ENGINEERING DESIGN & SOCIETY. 0-6 Semester Hr.**
(I, II, S) Pilot course or special topics course. Topics chosen from special interests of instructor(s) and student(s). Usually the course is offered only once, but no more than twice for the same course content. Prerequisite: none. Variable credit: 0 to 6 credit hours. Repeatable for credit under different titles.

**EDNS598. SPECIAL TOPICS. 0-6 Semester Hr.**

**EDNS598. SPECIAL TOPICS. 0-6 Semester Hr.**

**EDNS599. INDEPENDENT STUDY. 0.5-6 Semester Hr.**
Individual research or special problem projects supervised by a faculty member, also, when a student and instructor agree on a subject matter, content, and credit hours. Variable credit: 0.5 to 6 credit hours. Repeatable for credit under different topics/experience and maximums vary by department. Contact the Department for credit limits toward the degree. Independent Study form must be completed and submitted to the Registrar.
**Course Learning Outcomes**

**EDNS707. GRADUATE THESES. 1-15 Semester Hr.**

## Program Director

Richard Krahenbuhl, Associate Research Professor, Geophysics

## Associate Program Director

Elizabeth Reddy, Assistant Professor, Engineering, Design & Society

## Applied Mathematics & Statistics

HES Data Science Track Representative

Dorit Hammerling, Professor

## Civil & Environmental Engineering

HES Environmental Track Representatives

Alina Handorean, Teaching Professor (joint appointment: EDS/CEE)